

Generated MCQs

Question 1 (Graph Theory – nan | LLM difficulty: 2)

A simple graph G has 5 vertices and 6 edges. How many edges does its complementary graph $\overline{G}$ have?

- A. 4edges
- B. 5edges
- C. 10edges
- D. 6edges

Question 2 (Graph Theory – nan | LLM difficulty: 3)

Is the complete bipartite graph $K_{3,3}$ planar?

- A. No, it contains a subgraph homeomorphic to $K_{3,3}$, which is nonplanar by Kuratowski's theorem.
- B. *Yes, it is bipartite and bipartite graphs are always planar.*
- C. *Yes, it has no odd cycles and thus satisfies planarity conditions.*
- D. *Yes, it is a tree and all trees are planar.*

Question 3 (Graph Theory – nan | LLM difficulty: 3)

Consider a graph with vertices A, B, C, D and edges AB, BC, CD, DA, AC . Which of the following sequences is a path in this graph?

- A. $A \rightarrow B \rightarrow C \rightarrow D$
- B. $A \rightarrow B \rightarrow C \rightarrow A$
- C. $A \rightarrow B \rightarrow A \rightarrow C$
- D. $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$

Question 4 (Graph Theory – nan | LLM difficulty: 3)

Consider the sequence $v_1 - e_1 - v_2 - e_2 - v_3 - e_3 - v_2 - e_4 - v_1$ in a simple graph. Which of the following best describes this sequence?

- A. *A walk but not a trail*
- B. *A path*
- C. *A circuit*
- D. *A trail*

Question 5 (Graph Theory – nan | LLM difficulty: 2)

For a simple graph G with n vertices and m edges, the number of edges in its complement graph $\overline{G}$ is:

- A. $\binom{n}{2} - m$
- B. m
- C. $\binom{n}{2} + m$
- D. $\binom{n}{2} - 2m$

Question 6 (Graph Theory – nan | LLM difficulty: 2)

What is the number of edges in a spanning tree of a connected graph with 10 vertices?

- A. 9
- B. 8
- C. 10
- D. 11

Question 7 (Graph Theory – nan | LLM difficulty: 3)

A connected planar graph has 8 vertices and 12 edges. How many regions does it divide the plane into?

- A. 6
- B. 5
- C. 7
- D. 8

Question 8 (Graph Theory – nan | LLM difficulty: 1)

Which of the following is a necessary condition for two graphs G and G' to be isomorphic?

- A. There exist one-to-one correspondences $f : V(G) \rightarrow V(G')$ and $g : E(G) \rightarrow E(G')$ preserving adjacency.
- B. *Both graphs have the same number of vertices.*
- C. *Both graphs have the same number of edges.*
- D. *Both graphs have the same degree sequence.*

Question 9 (Graph Theory – nan | LLM difficulty: 1)

Which of the following is true for a graph G and its complement $\overline{G}$?

- A. *They have the same number of vertices.*
- B. *They have the same number of edges.*
- C. *They are always isomorphic.*
- D. *They have identical degree sequences.*

Question 10 (Graph Theory – nan | LLM difficulty: 2)

How many edges does the complete bipartite graph $K_{2,3}$ have?

- A. 6
- B. 5
- C. 7
- D. 8

Question 11 (Graph Theory – nan | LLM difficulty: 1)

A simple graph G has 6 vertices and 10 edges. How many edges does its complement have?
(A) 5 (B) 10 (C) 15 (D) 6

- A. 5
- B. 10
- C. 15
- D. 6

Question 12 (Graph Theory – nan | LLM difficulty: 1)

What is the number of edges in the complement of a simple graph with 4 vertices and 2 edges?

- A. 4
- B. 2
- C. 3
- D. 6

Question 13 (Graph Theory – nan | LLM difficulty: 1)

Which of the following is true about the complement of a complete graph K_n ?

- A. *The complement is an empty graph with no edges.*
- B. *The complement is also a complete graph.*
- C. The complement has the same number of edges as K_n .
- D. The complement is a cycle graph C_n .

Question 14 (Graph Theory – nan | LLM difficulty: 3)

Which of the following must be true for a self-complementary graph G ?

- A. *The number of vertices is congruent to 0 or 1 modulo 4.*
- B. *The graph is regular.*
- C. *The graph has an even number of edges.*
- D. *The graph is bipartite.*

Question 15 (Graph Theory – nan | LLM difficulty: 3)

Which of the following properties is preserved under graph isomorphism?
(a) The specific labels of the vertices
(b) The adjacency matrix entries
(c) The degree sequence of the graph
(d) The planar embedding of the graph

- A. *(c) The degree sequence of the graph*
- B. *(a) The specific labels of the vertices*
- C. *(b) The adjacency matrix entries*
- D. *(d) The planar embedding of the graph*

Question 16 (Graph Theory – nan | LLM difficulty: 3)

What is the number of edges in the simple graph represented by the adjacency matrix $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$?

- A. 5
- B. 10
- C. 4
- D. 6

Question 17 (Graph Theory – nan | LLM difficulty: 3)

Given the adjacency matrix of a simple graph G :

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

, which of the following is the adjacency matrix of its complement $\overline{G}$?

A. $\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

B. $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$

C. $\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$

D. $\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

Question 18 (Graph Theory – nan | LLM difficulty: 3)

Consider the directed graph with vertices 1, 2, 3 and edges $1 \rightarrow 2$, $2 \rightarrow 3$, and $3 \rightarrow 1$. Which of the following is its incidence matrix?

A. $\begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$

B. $\begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$

C. $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$

D. $\begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$

Question 19 (Graph Theory – nan | LLM difficulty: 3)

How many edges does the complement of the cycle graph C_4 have?

- A. 2
- B. 3
- C. 4
- D. 6

Question 20 (Graph Theory – nan | LLM difficulty: 3)

Consider the directed graph with vertices v_1, v_2, v_3, v_4 and edges $a_1 : v_1ov_2$, $a_2 : v_2ov_3$, $a_3 : v_3ov_4$, $a_4 : v_4ov_1$, and $a_5 : v_2ov_4$. Which of the following is the correct incidence matrix of this directed graph?

A.
$$\begin{pmatrix} -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 \end{pmatrix}$$

B.
$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

C.
$$\begin{pmatrix} -1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 \end{pmatrix}$$

D.
$$\begin{pmatrix} -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Question 21 (Graph Theory – nan | LLM difficulty: 3)

What is the chromatic number of a cycle graph with 5 vertices?

- A. 3
- B. 2
- C. 4
- D. 1

Question 22 (Graph Theory – nan | LLM difficulty: 2)

What is the number of edges in the complete graph K_5 ?

- A. $\frac{5 \times 4}{2} = 10$
- B. $5 \times 4 = 20$
- C. $5^2 = 25$
- D. $5 + 4 = 9$

Question 23 (Graph Theory – nan | LLM difficulty: 2)

Which of the following is not equivalent to the others for a graph G ?

- A. G has exactly one cycle
- B. G is a tree
- C. G is connected and acyclic
- D. G has $n - 1$ edges

Question 24 (Graph Theory – nan | LLM difficulty: 2)

What is the number of edges in the complement of a graph G with 10 vertices and 25 edges?

- A. 20
- B. 25
- C. 45
- D. 90

Question 25 (Graph Theory – nan | LLM difficulty: 2)

Which of the following statements about trees is false?

- A. *A tree can have cycles.*
- B. *A tree is connected and acyclic.*
- C. A tree with n vertices has $n - 1$ edges.
- D. *There is exactly one path between any two vertices in a tree.*

Question 26 (Graph Theory – nan | LLM difficulty: 3)

Which of the following is a trail but not a path in a graph?

- A. *A walk with no repeated edges but with at least one repeated vertex.*
- B. *A closed walk with no repeated edges.*
- C. *A walk with no repeated vertices.*
- D. *A closed walk with no repeated vertices except the starting/ending one.*

Question 27 (Graph Theory – nan | LLM difficulty: 2)

Which of the following complete graphs is bipartite?

- A. K_2
- B. K_3
- C. K_4
- D. K_5

Question 28 (Graph Theory – nan | LLM difficulty: 3)

Consider the undirected graph with vertices 1, 2, 3, 4 and edges $\{1-2, 1-3, 2-4, 3-4\}$. Which of the following is its adjacency matrix?

A. $\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$

B. $\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$

C. $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$

D. $\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$

Question 29 (Graph Theory – nan | LLM difficulty: 2)

If G is a simple graph with 4 vertices and 2 edges, how many edges does its complement $\overline{G}$ have?

- A. ☐ 4
- B. ☐ 2
- C. ☐ 6
- D. ☐ 3

Question 30 (Graph Theory – nan | LLM difficulty: 2)

A connected planar graph has 7 vertices and 11 edges. How many faces does it have?

- A. $|F| = 6$
- B. $|F| = 5$
- C. $|F| = 7$
- D. $|F| = 8$

Question 31 (Graph Theory – nan | LLM difficulty: 3)

What is the maximum number of edges in a connected planar simple graph with 7 vertices?

- A. 15
- B. 14
- C. 16
- D. 12

Question 32 (Graph Theory – nan | LLM difficulty: 3)

What is the maximum number of edges in a simple planar graph with 6 vertices?

- A. 12
- B. 8
- C. 15
- D. 9

Question 33 (Graph Theory – nan | LLM difficulty: 3)

Consider the adjacency matrix of a simple graph G given below. What is the chromatic number of G ?

- A. $\chi(G) = 3$
- B. $\chi(G) = 2$
- C. $\chi(G) = 4$
- D. $\chi(G) = 5$

Question 34 (Graph Theory – nan | LLM difficulty: 2)

What is the chromatic number of the complete bipartite graph $K_{2,3}$?

- A. 2
- B. 1
- C. 3
- D. 4

Question 35 (Graph Theory – nan | LLM difficulty: 2)

What is the chromatic number of the cycle graph C_5 ?

- A. 3
- B. 2
- C. 4
- D. 5

Question 36 (Graph Theory – nan | LLM difficulty: 3)

Which of the following statements is true about the cycle graph C_7 ?

- A. *It is not bipartite because it contains an odd – length cycle.*
- B. *It is bipartite because it can be divided into two sets with no edges within each set.*
- C. *It is 3 – regular and bipartite.*
- D. *It has a chromatic number of 3.*

Question 37 (Graph Theory – nan | LLM difficulty: 4)

Which of the following statements is true for a connected planar simple graph with $v \geq 3$ vertices?

- A. The number of edges is at most $3v - 6$.
- B. The number of edges is at least $3v - 6$.
- C. The number of edges is at most $2v - 4$.
- D. The number of edges is at least $2v - 4$.

Question 38 (Graph Theory – nan | LLM difficulty: 2)

Consider a tree with 7 vertices. Which of the following statements are true?

i. It has exactly 6 edges. ii. It contains exactly one cycle. iii. It has at least two leaves.

- A. *Only statements i. and iii. are true.*
- B. *Only statements i. and ii. are true.*
- C. *Only statements ii. and iii. are true.*
- D. *All statements are false.*

Question 39 (Graph Theory – nan | LLM difficulty: 2)

Consider the following weighted graph with vertices A, B, C, D, E and edges with weights: AB=1, BE=2, AC=3, CE=1, AD=4, DE=1. Which path from A to E has the least total weight?

- A. The path ABE with total weight 3.
- B. The path ACE with total weight 4.
- C. The path ADE with total weight 5.
- D. The path $ABCE$ with total weight 5.

Question 40 (Graph Theory – nan | LLM difficulty: 3)

Consider the following weighted graph with edges AB (2), AC (5), BC (1), BD (3), CE (4), and DE (1). What is the shortest path from A to E?

- A. $A \rightarrow B \rightarrow D \rightarrow E$
- B. $A \rightarrow B \rightarrow C \rightarrow E$
- C. $A \rightarrow C \rightarrow E$
- D. $A \rightarrow B \rightarrow D \rightarrow C \rightarrow E$

Question 41 (Graph Theory – nan | LLM difficulty: 3)

A connected graph has 10 vertices and 15 edges. What is the minimum number of edges that must be removed to leave a spanning tree?

- A. 6
- B. 5
- C. 7
- D. 8

Question 42 (Graph Theory – nan | LLM difficulty: 3)

Which of the following is a spanning tree of the connected graph with vertices A, B, C, D, E and edges $AB, AC, AD, AE, BC, BD, BE, CD, CE$?

- A. AB, AC, AD, AE
- B. AB, BC, CD, DE, EA
- C. AB, BC, CD
- D. AB, AC, AD, AE, BC

Question 43 (Graph Theory – nan | LLM difficulty: 4)

Which of the following statements is equivalent to a graph being a tree? Consider a graph G with n vertices.

- A. (c) G is connected and has $n - 1$ edges
- B. (a) G is acyclic and has n edges
- C. (b) G is connected and has $n + 1$ edges
- D. (d) G contains exactly one cycle

Question 44 (Graph Theory – nan | LLM difficulty: 4)

Which of the following is a valid characterization of a tree?(A) A connected graph with n vertices and n edges.(B) An acyclic graph with n vertices and $n - 2$ edges.(C) A connected graph with n vertices and $n - 1$ edges.(D) A graph with n vertices, $n - 1$ edges, and exactly one cycle.

- A. (C) A connected graph with n vertices and $n - 1$ edges.
- B. (A) A connected graph with n vertices and n edges.
- C. (B) An acyclic graph with n vertices and $n - 2$ edges.
- D. (D) A graph with n vertices, $n - 1$ edges, and exactly one cycle.

Question 45 (Graph Theory – nan | LLM difficulty: 4)

What is the total weight of the minimum spanning tree for the graph with vertices A, B, C, D and edges $AB = 1, AC = 3, AD = 4, BC = 2, BD = 5, CD = 1$?

- A. 4
- B. 5
- C. 3
- D. 6

Question 46 (Graph Theory – nan | LLM difficulty: 3)

Consider the weighted graph with vertices A, B, C, D, E and edges with the following weights: $AB = 1, AC = 4, BC = 2, BD = 5, CD = 3, DE = 1, CE = 6$. Using Kruskal's algorithm, which of the following is the correct minimum spanning tree?

- A. $\{AB, DE, BC, CD\}$
- B. $\{AB, DE, BC, BD\}$
- C. $\{AB, DE, AC, CD\}$
- D. $\{AB, DE, BC, CE\}$

Question 47 (Graph Theory – nan | LLM difficulty: 1)

What is the number of edges in the complementary graph $\overline{G}$ of a simple graph G with n vertices and m edges?

- A. $\frac{n(n-1)}{2} - m$
- B. $\frac{n(n-1)}{2} + m$
- C. m
- D. $n(n-1)$

Question 48 (Graph Theory – nan | LLM difficulty: 3)

A connected planar graph has 10 vertices and 15 edges. How many regions does it have?

- A. 7
- B. 6
- C. 8
- D. 5

Question 49 (Graph Theory – nan | LLM difficulty: 4)

Let G be a simple graph with vertex set $[5] = \{1, 2, 3, 4, 5\}$ and edges defined by $\{a, b\} \in E$ if and only if $a \equiv b \pmod{2}$. Consider the following statements about the complementary graph $\overline{G}$:

- $\overline{G}$ is bipartite. - $\overline{G}$ has exactly 5 edges. - $\overline{G}$ is a complete graph. - $\overline{G}$ is disconnected.

Choose the correct option:

- A. *Only one of the sentences is correct.*
- B. *There are exactly two correct sentences.*
- C. *There are exactly three correct sentences.*
- D. *All the sentences are true.*

Question 50 (Graph Theory – nan | LLM difficulty: 2)

What is the chromatic number of a bipartite graph?

- A. 2
- B. 1
- C. 3
- D. 4

Question 51 (Graph Theory – nan | LLM difficulty: 2)

Determine the chromatic number of the following graph: a cycle with 5 vertices labeled A, B, C, D, E connected sequentially in a loop. What is the minimum number of colors needed to color the vertices so that no two adjacent vertices share the same color?

- A. 3
- B. 2
- C. 4
- D. 5

Question 52 (Graph Theory – nan | LLM difficulty: 2)

In which of the following graphs can we find an Eulerian trail (a trail on G that includes every edge of G such that the only repeated vertices are the start and the end, and the start and end may be different)?

$$\psi_1 = \begin{pmatrix} A & B & C & D & E & F \\ 1, 2 & \{1, 2\} & \{2, 3\} & \{3, 4\} & \{4, 5\} & \{5, 6\} \end{pmatrix}$$

$$\psi_2 = \begin{pmatrix} X & Y & Z & W & V & U \\ 1, 2 & \{1, 2\} & \{2, 3\} & \{3, 4\} & \{4, 5\} & \{5, 6\} \end{pmatrix}$$

$$\psi_3 = \begin{pmatrix} P & Q & R & S & T & U \\ 1, 2 & \{1, 2\} & \{2, 3\} & \{3, 4\} & \{4, 5\} & \{5, 6\} \end{pmatrix}$$

$$\psi_4 = \begin{pmatrix} M & N & O & P & Q & R \\ 1, 2 & \{1, 2\} & \{2, 3\} & \{3, 4\} & \{4, 5\} & \{5, 6\} \end{pmatrix}$$

$$\psi_5 = \begin{pmatrix} L & M & N & O & P & Q \\ 1, 2 & \{1, 2\} & \{2, 3\} & \{3, 4\} & \{4, 5\} & \{5, 6\} \end{pmatrix}$$

- A. Graph ψ_1
- B. Graphs ψ_2 and ψ_3
- C. *None of them has an Eulerian trail.*
- D. *All of them have an Eulerian trail.*

Question 53 (Graph Theory – nan | LLM difficulty: 3)

What is the number of regions in a connected planar graph with 6 vertices and 9 edges?

- A. 5
- B. 4
- C. 6
- D. 7

Question 54 (Graph Theory – nan | LLM difficulty: 3)

What is the number of edges in a simple undirected graph with the following adjacency matrix?

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

- A. 5
- B. 10
- C. 4
- D. 6

Question 55 (Graph Theory – nan | LLM difficulty: 3)

Consider the undirected graph G with the following incidence matrix:

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

. How many connected components does G have?

- A. 2
- B. 1
- C. 3
- D. 4

Question 56 (Graph Theory – nan | LLM difficulty: 1)

Which of the following correctly describes the complementary graph of a simple graph G ?

- A. Two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G .
- B. Two vertices are adjacent in $\overline{G}$ if they are adjacent in G .
- C. The complement has the same edges as G but different vertices.
- D. *The complement has no edges.*

Question 57 (Graph Theory – nan | LLM difficulty: 3)

Which adjacency matrix represents the complementary graph of the graph with edges $\{1-2, 1-3, 2-3\}$ on four vertices?

A. $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

Question 58 (Graph Theory – nan | LLM difficulty: 3)

Consider a simple graph G with 5 vertices. If the degree sequence of G is $[2, 2, 2, 2, 2]$, what is the degree sequence of its complement graph $\overline{G}$?

- A. $[2, 2, 2, 2, 2]$
- B. $[3, 3, 3, 3, 3]$
- C. $[1, 1, 1, 1, 1]$
- D. $[0, 0, 0, 0, 0]$

Question 59 (Graph Theory – nan | LLM difficulty: 3)

Consider the adjacency matrix $A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$ of a simple graph G . Which of the following statements is true about G ?

- A. (a) G contains a cycle of length 3
- B. (b) G is bipartite
- C. (c) G has exactly 4 edges
- D. (d) G is a complete graph

Question 60 (Graph Theory – nan | LLM difficulty: 3)

Which of the following graphs is the complement of the graph G with vertex set $\{1, 2, 3, 4\}$ and edge set $\{\{1, 2\}, \{1, 3\}, \{1, 4\}\}$?

- A. (b) $\{\{2, 3\}, \{2, 4\}, \{3, 4\}\}$
- B. (a) $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$
- C. (c) $\{\{1, 2\}, \{1, 4\}, \{2, 4\}\}$
- D. (d) $\{\{1, 3\}, \{2, 3\}, \{3, 4\}\}$

Question 61 (Graph Theory – nan | LLM difficulty: 4)

Apply Kruskal's algorithm to find the minimum spanning tree of the weighted graph with vertices A, B, C, D and edges $AB = 1, AC = 2, AD = 3, BC = 4, BD = 5, CD = 6$. What is the correct sequence of edges selected?

- A. AB, AC, AD
- B. AB, AC, BC
- C. AB, AD, BD
- D. AB, AC, CD

Question 62 (Graph Theory – nan | LLM difficulty: 4)

Determine the chromatic number of the graph with vertices A, B, C, D and edges AB, BC, CA, AD, BD, CD using the described vertex coloring procedure. The graph consists of a triangle ABC with a central vertex D connected to all three vertices of the triangle.

- A. ☐ 4
- B. ☐ 3
- C. ☐ 2
- D. ☐ 5

Question 63 (Graph Theory – nan | LLM difficulty: 3)

How many spanning trees does the complete graph K_4 have?

- A. 16
- B. 8
- C. 12
- D. 24

Question 64 (Graph Theory – nan | LLM difficulty: 3)

Determine the chromatic number of the graph consisting of a 5-node cycle with an additional central node connected to all nodes of the cycle.

- A. 4
- B. 3
- C. 2
- D. 5

Question 65 (Graph Theory – nan | LLM difficulty: 3)

Consider the directed graph with vertices A, B, C, D and edges $A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A, A \rightarrow C$. What is its incidence matrix?

A.
$$\begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

B.
$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

C.
$$\begin{bmatrix} 1 & 0 & 0 & -1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

D.
$$\begin{bmatrix} 1 & 0 & 0 & -1 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix}$$

Question 66 (Graph Theory – nan | LLM difficulty: 3)

Which of the following is true about a simple connected graph with 6 vertices and 15 edges?

- A. *It is nonplanar.*
- B. *It is planar because it satisfies Euler's formula.*
- C. *It is a tree since it has no cycles.*
- D. *It is bipartite because it contains no odd-length cycles.*

Question 67 (Graph Theory – nan | LLM difficulty: 3)

Which of the following statements is true for any non-trivial tree? (A) It has exactly two leaves. (B) It has at least two leaves. (C) It has no leaves. (D) It has exactly one leaf.

- A. *It has at least two leaves.*
- B. *It has exactly two leaves.*
- C. *It has no leaves.*
- D. *It has exactly one leaf.*

Question 68 (Graph Theory – nan | LLM difficulty: 1)

A connected planar graph has 6 vertices and 9 edges. How many regions does it have in its planar representation?

- A. 5
- B. 13
- C. 4
- D. 7

Question 69 (Graph Theory – nan | LLM difficulty: 1)

A simple graph whose vertex set can be partitioned into two disjoint subsets such that every edge connects a vertex from one subset to a vertex in the other is called a:

- A. *bipartitegraph*
- B. *completegraph*
- C. *regulargraph*
- D. *disconnectedgraph*

Question 70 (Graph Theory – nan | LLM difficulty: 3)

Which inequality must hold for a simple connected planar graph with $v \geq 3$ vertices and e edges?

- A. $e \leq 3v - 6$
- B. $e \leq 2v - 4$
- C. $e \leq 3v - 5$
- D. $e \leq 3v - 3$